

Pneumonia – Summary

Key Symptoms: Fever, chills, chest pain, productive cough, high HR, headaches, myalgia, N/V, SOB, etc.

Most common causative pathogens

- **Community-Acquired Pneumonia (CAP)**
 - **Outpatient:** *S. pneumo* (most common), *M. catarrhalis*, *C. pneumo*, *M. pneumo*, *S. aureus*, *H. influenzae*
 - **Inpatient (non-ICU):** Outpatient pathogens, +/- gram-negative bacilli
 - **Inpatient (ICU):** Outpatient pathogens +/- MRSA, *Legionella* spp., *Enterobacteriales*
- **Hospital-Acquired/Ventilator-Associated Pneumonia (HAP/VAP):** *S. aureus* (incl. MRSA), *P. aeruginosa*, Enteric gram-negative bacilli, *S. pneumo* (less common)

Diagnosis: X-ray, CT, MRI, culture and sensitivities. Scoring systems like PSI and CURB-65, see slide 7.

Empiric Pharmacological Treatment

- **CAP:** Typically, 5–7-day treatment
 - **Outpatient:** Amoxicillin (preferred) **OR** doxycycline **OR** macrolides (least preferred due to increasing resistance)
 - **Outpatient (risk factors):** (Amoxi-clav **OR** 2nd gen cephalosporin **PLUS** (macrolide **OR** doxycycline)) **OR** Respiratory fluoroquinolone (e.g. moxifloxacin, levofloxacin)
 - **Inpatient (non-ICU):** Same as outpatient with 3rd gen cephalosporin instead of 2nd gen and macrolide preferred over doxycycline
 - **Inpatient (ICU):** IV beta-lactam **PLUS** (IV fluoroquinolone **OR** IV azithromycin)
 - **Inpatient (suspected pseudomonas):** (Antipseudomonal beta-lactam **PLUS** ciprofloxacin **OR** levofloxacin) **OR** (Antipseudomonal beta-lactam **PLUS** aminoglycoside **PLUS** IV azithromycin)
 - **Antipseudomonal beta-lactams:** Cefepime, ceftazidime, piperacillin-tazobactam, carbapenems (except ertapenem)
 - **HAP/VAP:** Typically, 7-day treatment but may be up to 10-14 days in certain cases
 - **No MRSA:** Pip/tazo, cefepime, fluoroquinolones, meropenem, imipenem, aminoglycosides (unstable/high risk patients)
 - **If MRSA suspected:** vancomycin, linezolid
- ❖ See slides 12 & 14 for targeted therapies (once causative pathogen has been identified)

Monitoring Parameters: Majority of symptom improvement expected within 48 hours but resolution by 3-5 days (up to 1-2 weeks). Chest x-ray may take 4-6 weeks to normalize.

Additional Key Points

- For empiric therapy, administer an antibiotic from a different class if the patient has received Abx 3 months prior to CAP diagnosis
- Aminoglycosides do not penetrate lung tissue well, used for antipseudomonal properties and synergy when combined with beta-lactams, use alternative regimens if possible
- Daptomycin is **NOT** used for MRSA in pneumonia as it is inactivated by pulmonary surfactant
 - Daptomycin has a large lipopeptide structure susceptible to the surfactant
- Ciprofloxacin has poor *S. pneumoniae* coverage, but has coverage for pseudomonas